

1. Executive Summary

		Traffic Summary		Count			
		Total Flows Observed		18287			
Total Threats Mitigated		14		Total Packets Analyzed		1706050080	
High Priority		6		Malicious Packets		1704814744	
Low Priority		8		True Negatives Passed		13322	
Manual Releases		0					
Manual Blocks		0					
				Attack Vector		Count	
				ICMP Flood		0	
ML Metrics		Value		SYN Flood		0	
False Positives		38		UDP Flood		10	
FP Rate		0.29%		Uncertain		4	
IF Precision		99.23%					
IF Recall		98.95%		Action		Count	
IF F1-Score		99.09%		Quarantined		3	
IF Accuracy		99.51%		Time Ban		3	
				Blackhole		0	
				Released		1	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.23%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.95%	Share of actual attacks successfully flagged
F1-Score	99.09%	Balanced measure combining Precision and Recall
Accuracy	99.51%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.29%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.05%	Actual attacks that went undetected
True Positive Rate (TPR)	98.95%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.71%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4923	FN 52
Actual: Normal	FP 38	TN 13270

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4923

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.23%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.95%	Share of actual attacks successfully flagged
F1-Score	99.09%	Balanced measure combining Precision and Recall
Accuracy	99.51%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.29%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.05%	Actual attacks that went undetected
True Positive Rate (TPR)	98.95%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.71%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4923	FN 52
Actual: Normal	FP 38	TN 13270

2d. Response Latency

Metric	Value	Description
Detection Time	660.8 ms	Average time taken to detect an attack
Mitigation Response Time	0.2 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.13%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	7.15%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.03%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	4	3	2026-09-26 21:30:29	2026-09-26 21:33:10
10.0.0.21	4	3	2026-09-26 21:30:38	2026-09-26 21:33:14
10.0.0.22	4	3	2026-09-26 21:30:33	2026-09-26 21:33:11
10.0.0.1	1	1	2026-09-26 21:27:26	2026-09-26 21:27:37

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 21:27:37	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 21:30:29	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 21:30:33	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 21:30:38	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 21:30:40	10.0.0.20	DDoS	UDP Flood	76.1%	High	Time Ban (30s)
2026-09-26 21:30:40	10.0.0.20	DDoS	UDP Flood	76.1%	High	Time Ban
2026-09-26 21:30:44	10.0.0.22	DDoS	UDP Flood	86.8%	Critical	Time Ban (30s)
2026-09-26 21:30:45	10.0.0.22	DDoS	UDP Flood	86.8%	Critical	Time Ban
2026-09-26 21:30:49	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (30s)
2026-09-26 21:30:50	10.0.0.21	DDoS	UDP Flood	78.1%	Critical	Time Ban
2026-09-26 21:31:41	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban (1m)
2026-09-26 21:31:45	10.0.0.22	DDoS	UDP Flood	86.8%	High	Time Ban (1m)
2026-09-26 21:31:50	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban (1m)
2026-09-26 21:33:11	10.0.0.22	DDoS	UDP Flood	86.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6471	77.9%	Critical	Phase 2	1	11s	2026-09-26 21:33:14	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6329	86.0%	Critical	Phase 2	1	11s	2026-09-26 21:33:11	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	79.1%	Critical	Phase 2	1	11s	2026-09-26 21:33:10	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6293	79.6%	High	Phase 2	3	1m 12s	2026-09-26 21:31:50	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6252	86.8%	High	Phase 2	3	1m 12s	2026-09-26 21:31:45	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6284	79.4%	High	Phase 2	3	1m 12s	2026-09-26 21:31:41	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6456	79.6%	High	Phase 2	2	42s	2026-09-26 21:31:20	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6338	86.8%	Critical	Phase 2	2	42s	2026-09-26 21:31:15	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6452	79.4%	High	Phase 2	2	41s	2026-09-26 21:31:10	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	78.1%	High	Phase 2	1	11s	2026-09-26 21:30:49	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6338	86.8%	Critical	Phase 2	1	11s	2026-09-26 21:30:44	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	76.1%	High	Phase 2	1	11s	2026-09-26 21:30:40	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.6054	58.9%	Low	Phase 1	1	11s	2026-09-26 21:27:37	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE